



PSOC™ Edge designing a secure system

Customer training workshop (CTW)

April 2026



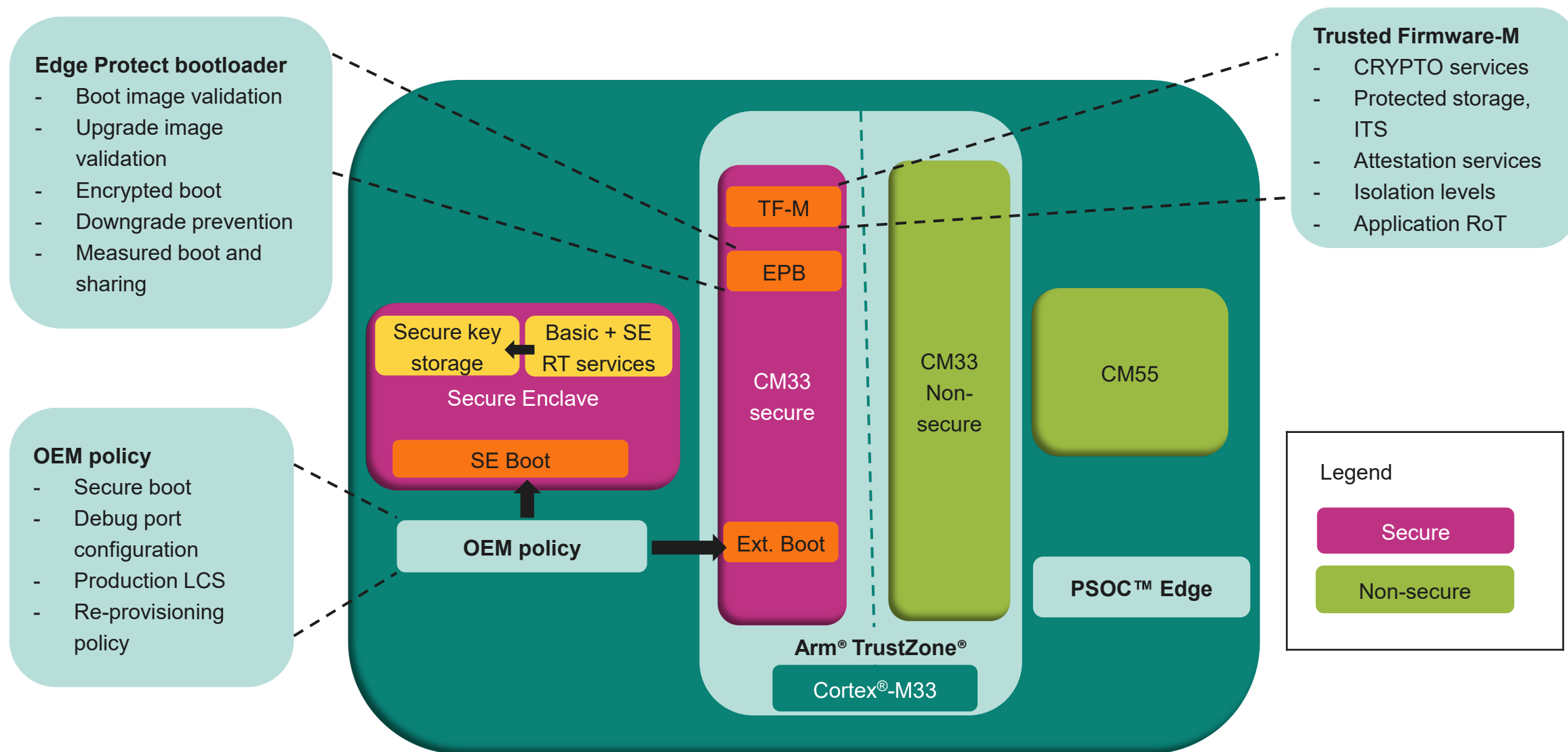
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Secure design overview (EPC 4)



Secure design overview (EPC 2)

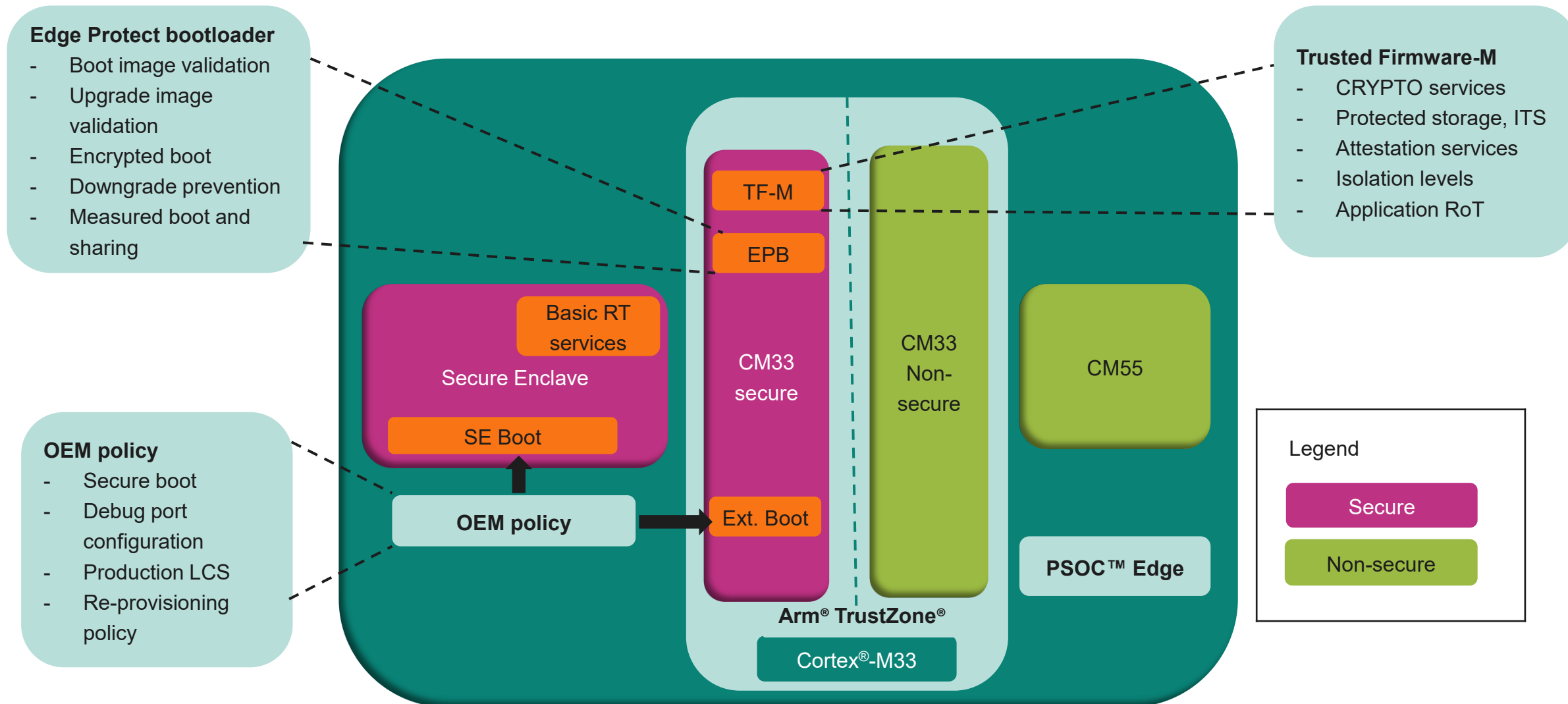


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Peripheral and memory protection controller (MPC and PPC)

- PSOC™ Edge memory and peripheral protection can be configured using the peripheral protection controller (PPC) and the memory protection controller (MPC) regions
- The PPC defines the access permissions of a peripheral resource, such as a serial communication block (SCB)
- The MPC applies similar protection to memory regions. It is akin to Arm® implementation-defined attribution unit (IDAU), but offers more granular control
- Together, the PPC and MPC enable a highly customizable security configuration tailored to a specific device
- MPC and PPC regions can be configured using the ModusToolbox™ device configurator

Note: device configurator protections are applied within the `cybsp_init()` call from the CM33 secure application.

Configure memory protections

- Choose appropriate domains for each memory region

The screenshot shows the 'Memory' tab in the Infineon Device Configurator 5.50. The 'Memory Regions' section lists various memory blocks. The 'm33s_nvm' region is highlighted, and the 'Edit Memory Region' dialog is open for it.

Memory Regions Table:

Region Id	Domain	Offset	Size	Description
UNALLOCATED	<NONE>	0x00000000	0x00100000 (1 MB)	
m33s_nvm	M33S_CODE (Domain 6)	0x00100000	0x00200000 (2 MB)	CM33 secure image
		0x00300000	0x00040000 (256 KB)	CM33 secure trailer
		0x00340000	0x00200000 (2 MB)	CM33 image
		0x00540000	0x00040000 (256 KB)	CM33 trailer
		0x00580000	0x00200000 (2.75 MB)	CM55 image
		0x00840000	0x00040000 (256 KB)	CM55 trailer
		0x00880000	0x00240000 (2.25 MB)	
		0x00AC0000	0x00240000 (2.25 MB)	

Edit Memory Region - Device Configurator 5.50

Region id: m33s_nvm

Description: CM33 secure image including non-secure callable region

Domain: M33S_CODE (Domain 6)

Offset: M33 (Domain 1)

Size (bytes): M33_M55 (Domain 4)

Addresses: M33NSC (Domain 3)

M33S (Domain 0)

M33S_CODE (Domain 6)

M55 (Domain 2)

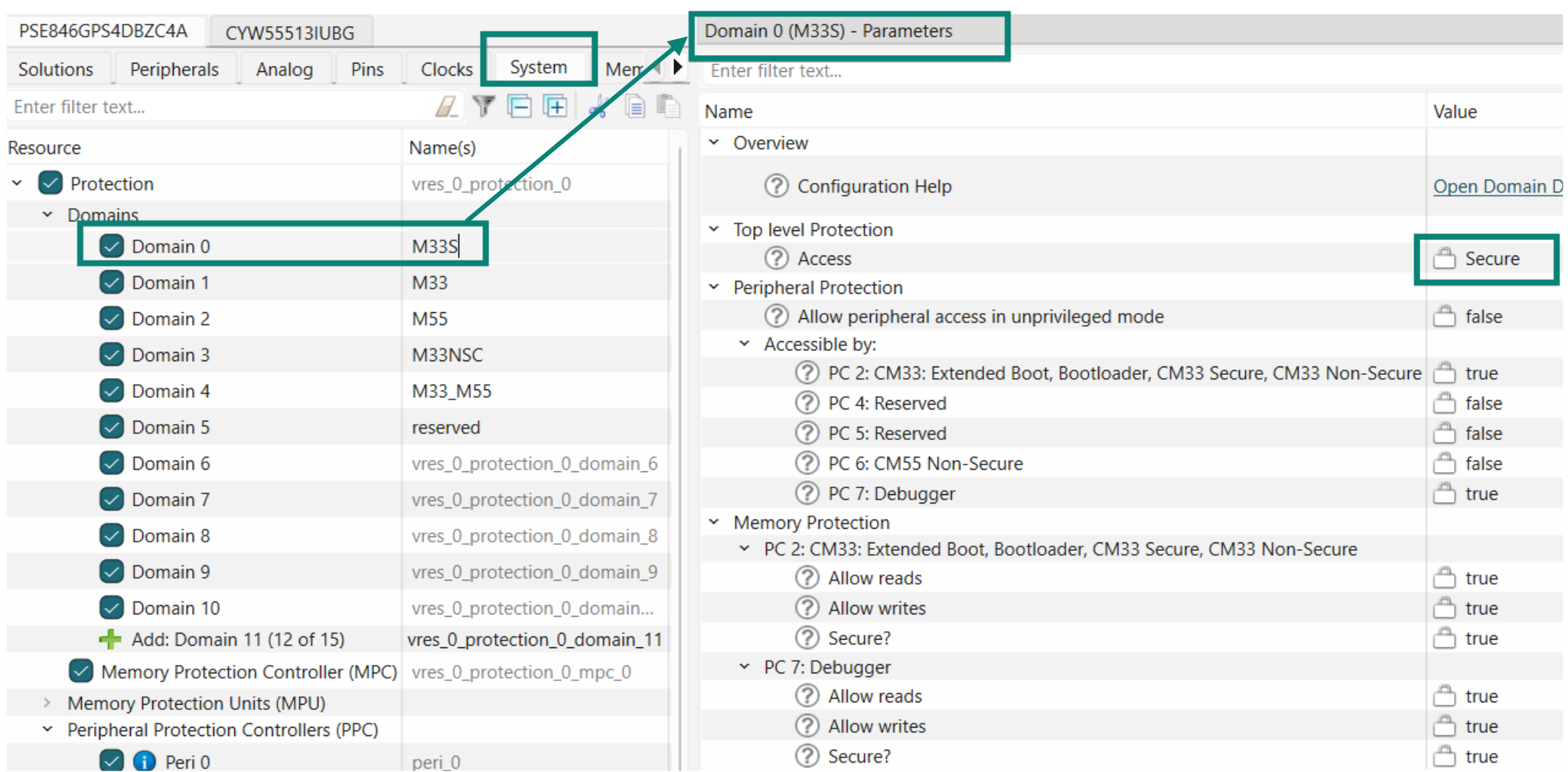
reserved (Domain 5)

TFM_SP_CRYPT0 (Domain 8)

TFM_SP_INITIAL_ATTESTATION (Domain 7)

TFM_SP_ITS (Domain 9)

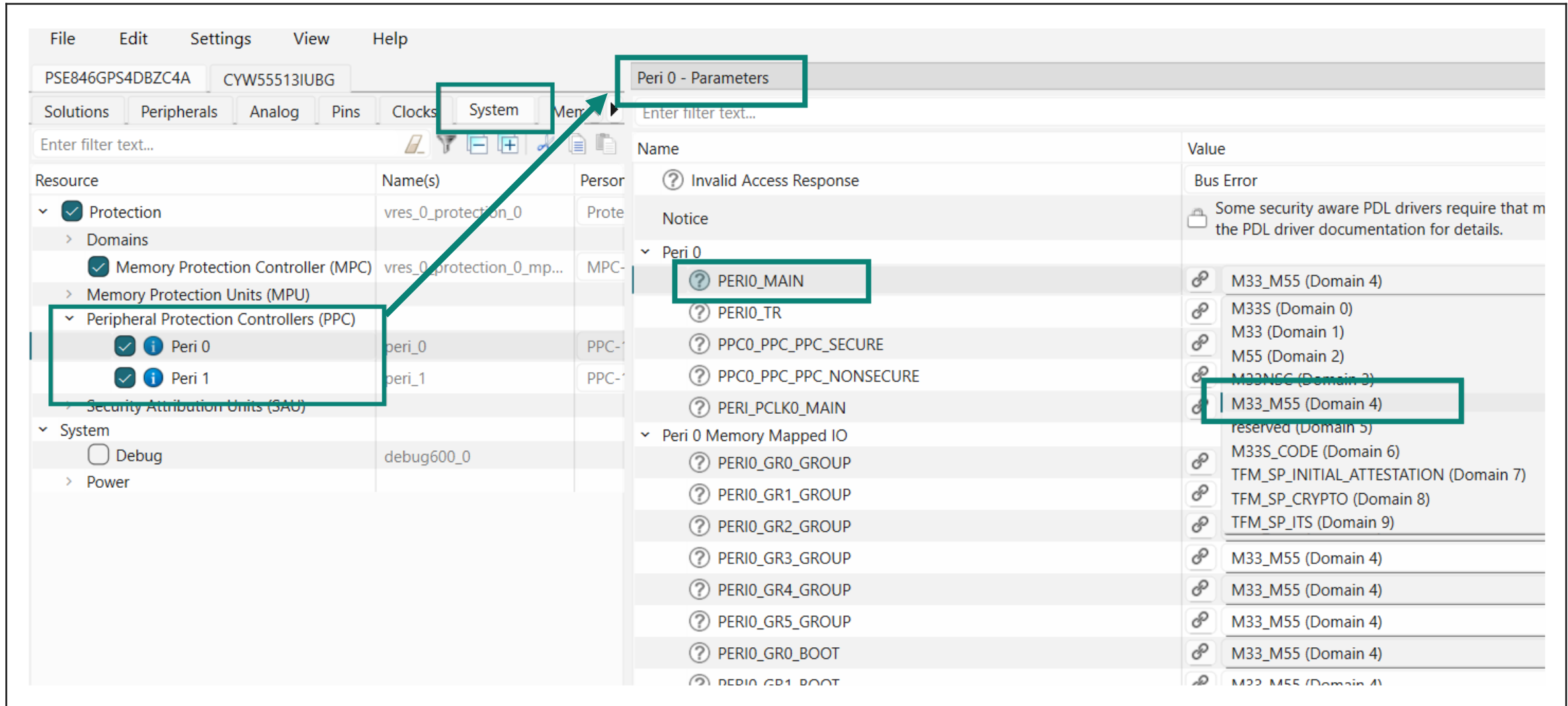
Domain parameters



The screenshot displays the Infineon development tool interface for configuring domain parameters. The 'System' tab is selected, and the 'Domain 0 (M33S) - Parameters' configuration is active. The left pane shows a tree of resources with 'Domain 0' selected. The right pane shows the configuration for 'Domain 0 (M33S) - Parameters'.

Name	Value
Overview	
Configuration Help	Open Domain D
Top level Protection	
Access	Secure
Peripheral Protection	
Allow peripheral access in unprivileged mode	false
Accessible by:	
PC 2: CM33: Extended Boot, Bootloader, CM33 Secure, CM33 Non-Secure	true
PC 4: Reserved	false
PC 5: Reserved	false
PC 6: CM55 Non-Secure	false
PC 7: Debugger	true
Memory Protection	
PC 2: CM33: Extended Boot, Bootloader, CM33 Secure, CM33 Non-Secure	
Allow reads	true
Allow writes	true
Secure?	true
PC 7: Debugger	
Allow reads	true
Allow writes	true
Secure?	true

Configure peripheral protections



The screenshot shows the Infineon development tool interface for configuring peripheral protections. The 'System' tab is selected, and the 'Peripheral Protection Controllers (PPC)' are expanded. 'Peri 0' is selected, and its parameters are shown in the right pane. A green arrow points from the 'Peri 0' selection in the left pane to the 'Peri 0 - Parameters' title in the right pane. The 'Peri 0 - Parameters' pane shows a list of parameters with their values. The 'PERI0_MAIN' parameter is highlighted, and its value 'M33_M55 (Domain 4)' is also highlighted in the right pane.

Name	Value
Invalid Access Response	Bus Error
Notice	Some security aware PDL drivers require that m the PDL driver documentation for details.
Peri 0	
PERI0_MAIN	M33_M55 (Domain 4)
PERI0_TR	M33S (Domain 0)
PPC0_PPC_PPC_SECURE	M33 (Domain 1)
PPC0_PPC_PPC_NONSECURE	M55 (Domain 2)
PERI_PCLK0_MAIN	M33NSC (Domain 3)
Peri 0 Memory Mapped IO	
PERI0_GR0_GROUP	M33_M55 (Domain 4)
PERI0_GR1_GROUP	M33S_CODE (Domain 6)
PERI0_GR2_GROUP	TFM_SP_INITIAL_ATTESTATION (Domain 7)
PERI0_GR3_GROUP	TFM_SP_CRYPTO (Domain 8)
PERI0_GR4_GROUP	TFM_SP_ITS (Domain 9)
PERI0_GR5_GROUP	M33_M55 (Domain 4)
PERI0_GR0_BOOT	M33_M55 (Domain 4)
PERI0_GR1_BOOT	M33_M55 (Domain 4)

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ModusToolbox™ secure request framework (SRF)

- The ModusToolbox™ secure request framework (SRF) provides a reusable infrastructure for implementing multiple secure operations via a single non-secure callable (NSC) entry point, which also performs basic argument validation
- It supports calling across the security barrier of a single core, and also calling from a non-TrustZone core to a Arm® TrustZone® core via an Interprocessor communication (IPC)
- It integrates Arm® Cortex®-M security extensions (CMSE) and Infineon's peripheral protection controller (PPC) and Memory Protection Controller (MPC) regions into a single streamlined process

Note: See the [SRF user guide](#) for more details.

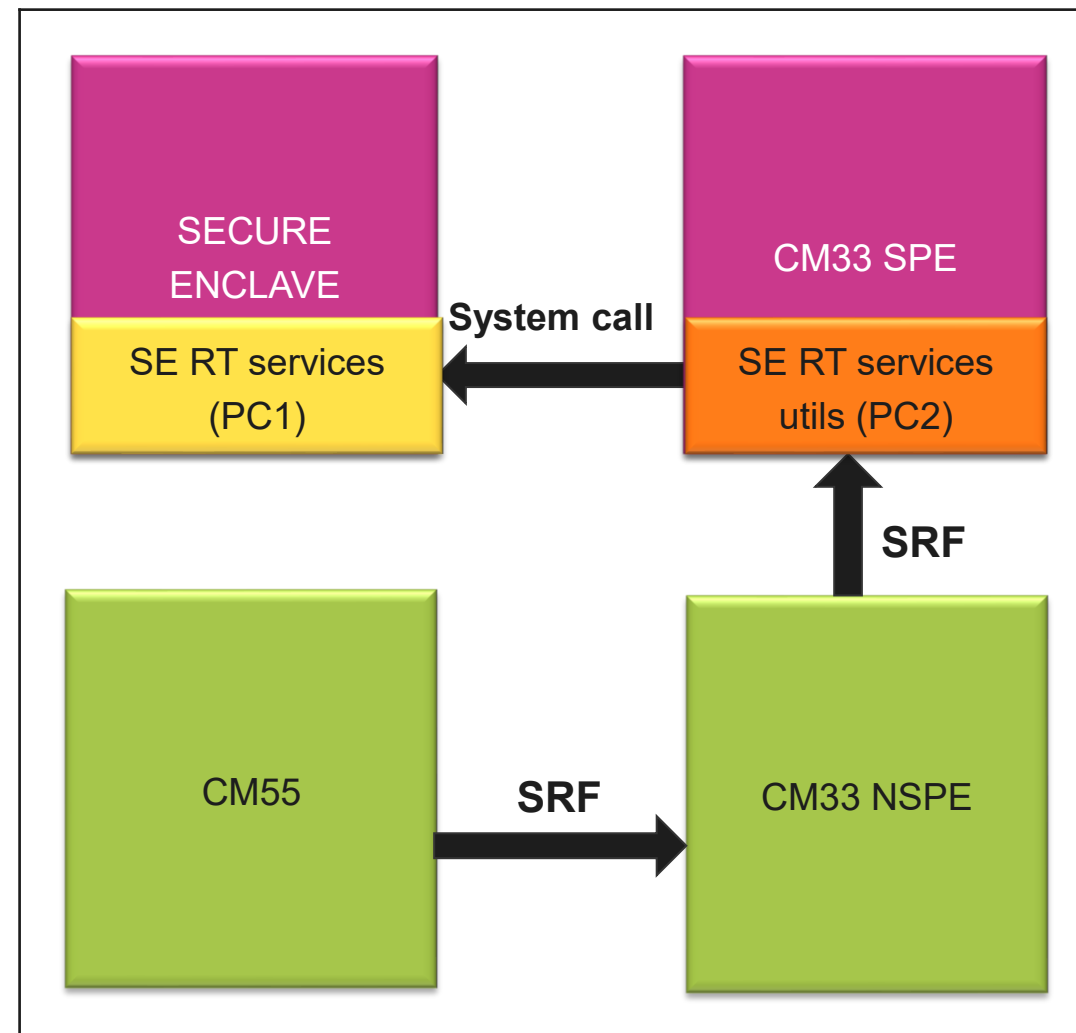
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Secure enclave runtime (SE RT) services (EPC 4)

- Edge Protect Category (EPC) 4 devices provide an additional set of services called secure enclave runtime services (SE RT_Services), which include
 - CRYPTO service
 - Key management service
 - Protection service
 - Attestation service
 - System management service
- SE RT services utilities library is a set of utility code that provides an interface to SE RT services
- SE RT services utilities can be accessed from the CM33 SPE application.

Note: Ensure peripherals are configured as secure. See training on [“Security tips and tricks”](#) for more details.



Secure enclave runtime (SE RT) services key storage (EPC 4)

SE RT services also provide special storage within the Secure Enclave to store persistent keys

User keys can be imported into the Secure Enclave in two ways:

1. During runtime using `ifx_se_import_key()` API
2. By providing the key in the “user keys” section in the policy file during provisioning

Note: See the [SE RT](#) Services Utilities Library for more details.

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Transition the device to production LCS

- Before the device is shipped out to the field, ensure that the device is in the production LCS
- Ensure that two OEM key pairs are generated and stored securely (recommended to use HSM for key generation and signing)

```
$ edgeprotecttools create-key --key-type ECDSA-P256 --output keys/oem_private_key_0.pem
keys/oem_public_key_0.pem
```

```
$ edgeprotecttools create-key --key-type ECDSA-P256 --output keys/oem_private_key_1.pem
keys/oem_public_key_1.pem
```

- OEM needs to generate the Certificate Signing Request (CSR) with the production type certificate

```
$ edgeprotecttools -t pse8xs2 oem-csr --certificate-name "oem-cert" --oem "Dummy OEM" --
project "Dummy Project" --project-number "1234" --public-key-0 keys/oem_public_key_0.pem
--public-key-1 keys/oem_public_key_1.pem --cert-type production --output packets/apps/
prov_oem/oem_csr_prod.bin --sign-key-0 keys/oem_private_key_0.pem --sign-key-1 keys/
oem_private_key_1.pem
```

- See the application note [Getting started with PSOC™ Edge security](#) for instructions to transition the device to Production LCS

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Re-provisioning

- In the production LCS, the device can only be re-provisioned. Provisioning is not allowed
- The OEM policy fields that need to be altered via re-provisioning must be configured as re-provisionable in the reprovisioning policy object of the OEM policy file. This OEM policy file must be used during the transition to Production LCS
- Only those fields in the OEM policy that were enabled for re-provisioning can be altered later via the re-provisioning process

Note: If the system access port is closed during the transition to Production LCS, then re-provisioning is not possible through the SWD interface. Alternatively, the user can close the ports using the re-provisioning step.

```
$ edgeprotecttools -t pse8xs2 reprovision-device -p policy/policy_oem_provisioning.json --key  
keys/oem_private_key_0.pem
```

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Debug ports configuration

- Configure the debug ports in the OEM policy file and provision the device
- Infineon recommends that all the debug ports be disabled for the final production units.
- Certificate-based authenticated debug mechanism is supported
- The device cannot be re-provisioned via SYS-AP once this port has been disabled in the Production LCS
- By design, the SYS-AP always remains open in the Development LCS to ensure the device can always be provisioned to the default state and does not get bricked due to incorrect policy configurations

```

"debug": {
  "cm33": {
    "cpu": {
      "description": "CPU access port debug restriction",
      "applicable_conf": "Disabled, Enabled, Allowed",
      "value": "Disabled"
    },
    "allowed": {
      "dbggen": {
        "niden": {
          "spiden": {
            "spniden": {
              "secure": {
            },
            "system": {
              "ap": {
                "description": "System access port (SYS-AP) restriction. The SYS
                "applicable_conf": "Disabled, Enabled, Allowed",
                "value": "Disabled"
              },
              "allowed": {
                "mpc_ppc_enable": {
                  "rram_otp_enable": {
                    "rram_nvm_enable": {
                      "ram_enable": {
                        "mmio_enable": {

```

Revision history

Document revision	Date	Description of change
**	2026-01-12	Initial release.
*A	2026-04-24	Updated speaker notes Updated images to align with brand guidelines

